ALI (FARZAD) JAMALI

AI & Computer Vision Engineer / MPhil Researcher

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Profile

I am an AI and Computer Vision engineer with research experience at the University of Salford. I build AI systems that work on real industrial and commercial data. My projects include extracting structured data from images and technical documents and detecting anomalies and defects in industrial images. I have also built deep learning based recommendation systems and intelligent analytics dashboards for World Privilege Plus. In my research, I work with Gen AI and LLMs, including their use, improvement and evaluation. My mechanical engineering background helps me connect these methods to real engineering problems.

Technical Skills

Computer Vision: segmentation, OCR, document intelligence, defect detection, image preprocessing, geometry extraction, leakage-safe evaluation *(OpenCV, Tesseract, U-Net, SegFormer)*.

Machine Learning: transfer learning, embeddings, confidence calibration, robustness testing, statistical evaluation, error analysis *(scikit-learn)*.

Deep Learning: model training and fine-tuning, vision-language model adaptation *(PyTorch, torchvision, Hugging Face Transformers)*.

LLMs & MLLMs: multimodal representation learning, retrieval-augmented generation (RAG), recommender systems, evidence-grounded explanation and verification, agentic workflows, LLM behaviour evaluation *(CLIP, SigLIP2, LangChain, ChromaDB, Ollama)*.

Software & Data: Python, SQL, Pandas, NumPy, FastAPI, REST APIs, DuckDB, Parquet, Docker, Git, Linux, Streamlit, pytest, uv.

Engineering: technical drawings, stress analysis *(AutoCAD Mechanical, SolidWorks, CATIA)*.

Applied AI & Research Projects

Floor Plan Document Intelligence / *Independent project (real client brief), 2026*

- Built a pipeline on CubiCasa5K that turns floor-plan images and PDF pages into structured building data. Combined SegFormer-B0 segmentation, Tesseract OCR and vector geometry to find rooms and openings and link them to text labels.
- Raised five-class mIoU from a classical baseline of 0.312 to 0.577 on held-out data. Added a Streamlit interface for reviewing results and confidence scores.

Recommendation & Analytics Platform / *Delivered to World Privilege Plus, 2025–2026*

- Built a deep learning based, semantic-aware recommendation and analytics platform independently for a rewards business with over 4 million members. Used Python and FastAPI for data processing, model training and real-time predictions, with DuckDB and Parquet for analytics.
- Turned business requirements into a working system, explained the results to non-technical colleagues and handed over the completed platform for use in the business.

Industrial Fabric Defect Inspection / *Independent project (real client brief), 2026*

- Built a defect segmentation pipeline on the AITEX dataset using PyTorch and OpenCV. Compared a classical texture baseline with a U-Net/ResNet-18 model and processed full-resolution images in tiles.
- Used a locked holdout set to control for data leakage. Dice rose from 0.259 to 0.532, with image-level ROC-AUC of 0.878.

Evidence-Constrained Multimodal Fashion Recommendation / *MPhil thesis project*

- Built a multimodal fashion recommender using CLIP image and text embeddings for similarity search, then reranked candidates using a 200-rule fashion knowledge base. Generated explanations with RAG using the exact rules behind each recommendation.
- Verified every explanation automatically. Extracted atomic claims and checked each one against the rule evidence, across 1,000 test cases and three open-source LLMs.
- Supplying the exact rule trace raised trace-supported claims by 21 percentage points (95% CI 19.7 to 22.4). Reported that reranking itself did not improve retrieval metrics.

LLM Output Homogenization / *NLP & LLM Evaluation*
(Manuscript in preparation for journal submission)

- Compared text generated by multiple LLMs across different datasets to study similarities and differences in their outputs.
- Used text embeddings and semantic similarity analysis to evaluate response diversity and consistency.

GitHub: LLM-output-homogenization

RepBench-Fashion / *Research paper (in preparation)*

- Built a controlled benchmark of seven frozen visual representations, including ResNet50, DINOv3, CLIP and SigLIP2, for fashion outfit compatibility. Tested across Polyvore, IQON3000 and a human-grounded external set, with no item or outfit overlap between training and test.
- Found that contrastive vision-language models beat ResNet50 and DINOv3 on both datasets (ROC-AUC about 0.89 against 0.69 on Polyvore). Fashion-specialised models were not consistently better. Used bootstrap intervals with Holm correction.

Teaching Experience

Hourly Paid Academic (HPA) – Deep Learning | University of Salford
Sep 2026 – Present | Salford, UK

- Run Deep Learning workshops for undergraduate and postgraduate students and supervise their work. Help them write and debug code in Python, TensorFlow and PyTorch.

Engineering & Data Experience

Data Analyst | Nilper
Sep 2023 – Sep 2025 | Iran

- Analysed sales, product and quality data across more than 20 product categories. Built dashboards and reports to support day-to-day business decisions.

Data Analyst | Solico Group
Sep 2021 – Mar 2023 | Iran

- Automated KPI reports for forecasting and performance tracking. Analysed consumer, campaign and product category data.

Mechanical Engineer | Padideh Sanat Ouj Co.
2019 | Iran

- Worked on hydrotest and hydroforming equipment, using SolidWorks and CATIA to design and analyse components including bolts, clamps and bearings.

Mechanical Design & Simulation Intern | SAIPA Corp.
2014 | Iran

- Produced orthographic and three-view drawings for manufacturing jigs and fixtures in AutoCAD Mechanical. Assisted with mechanical design and simulation.

Education

MPhil, Computer Science & Artificial Intelligence | University of Salford

Sep 2025 – Expected Nov 2026 | Salford, UK

- Thesis on evidence-grounded fashion recommendation. Built and evaluated a multimodal recommender that explains its choices and verifies those explanations against a knowledge base.

MSc Mechanical Engineering – Dynamics, Control & Vibrations | Shahid Beheshti University

2016 – 2019 | Tehran, Iran

- Researched solar tracking, mechanical design, modelling and control. Completed two engineering drawing modules and used CAD tools for design and stress analysis.

BSc Mechanical Engineering | Semnan University

2012 – 2016 | Semnan, Iran

Publication & Additional Information

Experimental evaluation of the open-loop control method for a dual-axis solar tracker. *Modares Mechanical Engineering*, 23(02), 127–138, 2023. DOI: 10.52547/mme.23.2.127

Mobility: Open to relocation within the UK.

Languages: Persian (native), English (full professional; IELTS 7.5).